**AARUSH DHINGRA**
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## Profile Summary

Computer Science undergraduate specializing in **full-stack development and scalable systems**, with hands-on experience building applications using modern technologies. Skilled in **LLM integration, prompt engineering, and API development**, with a focus on performance, modular design, and usability. Passionate about building efficient, high-impact software solutions.

## Education

**Bachelor of Technology – Computer Science & Data Science (2023 – 2027)**
CMR Institute of Technology, Bengaluru — CGPA: 8.71/10 (pursuing)

## Technical Skills

- **Languages:** Python, JavaScript, Java
- **Frontend:** React.js, Tailwind CSS
- **Backend:** Node.js, Express.js, REST APIs, JWT, OAuth 2.0, WebSockets
- **Databases:** MongoDB, MySQL, PostgreSQL, Redis, Supabase
- **Tools:** Docker, Git, GitHub, Postman
- **AI/LLM:** RAG, LangChain, LangGraph, Prompt Engineering, LLM Agents, Tool Calling, Embeddings, Vector Search
- **Core Fundamentals:** Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Object-Oriented Programming

## Internship Experience

**ThinkSeed Systems – Software Development Engineer Intern** July 2026 – January 2027 (ongoing)

- Developing a **test automation framework** and software-based phone simulator to automate hardware-dependent validation workflows, reducing manual testing effort by **60%**.
- Designing and implementing automated simulations of **real-device interactions and communication flows**, enabling repeatable execution of test scenarios and reducing test execution time by **40%**.

## Projects

***Prmpt — AI Prompt Optimizer*** *Electron.js, React.js, Rust, Python, OpenRouter API*

- Developed a cross-platform desktop application integrating LLMs to optimize AI prompts in real time via a system-wide global hotkey trigger, eliminating context switching for end users.
- Built a multi-stage NLP pipeline using Claude and GPT-4o via LiteLLM/OpenRouter for prompt classification, intent clarification, and structured output generation, streamed token-by-token over WebSocket.
- Built a modular prompt system (overlay + app) to manage reusable prompts, user context, and chat sessions.
- Designed and deployed a Retrieval-Augmented Generation (RAG) system for local codebase search, combining dense vector search (pgvector), full-text search, and neural sparse embeddings to retrieve relevant code context for LLM-augmented responses.

**NetShape - *Network Condition Simulator*** GitHub *Python, asyncio, Typer, JavaScript*

- Built and published an open-source CLI tool to PyPI that wraps any process in a local HTTP/HTTPS forward proxy, injecting bandwidth throttling, latency, packet loss, and jitter with zero code changes to the target app
- Implemented a token-bucket bandwidth throttling engine in Python `asyncio` with 12 built-in network profiles (2G through fiber) and real-time live adjustment without process restart
- Designed an embedded web dashboard with Server-Sent Events, Chart.js throughput/latency graphs, and interactive controls — served directly from the proxy with no external runtime dependencies
- Built per-endpoint throttle rules via regex pattern matching and YAML scenario scripting, enabling automated multi-phase network sequences (e.g. 4G → dead zone → 2G recovery)

**VidNova – Scalable Video Streaming Platform** GitHub *MongoDB, Express.js, React.js, Node.js*

- Built a full-stack video streaming platform with secure video hosting, playback, and user interaction features.
- Designed and implemented RESTful APIs with JWT authentication for secure and efficient client-server communication.
- Optimized MongoDB performance using indexing and schema design; developed a scalable channel ecosystem with subscriptions, likes, and comments.

## Achievements

- Published **NetShape**, an open-source network simulation tool on PyPI.
- Ranked **Top 5** in BytesBattle Hackathon (CMRIT)
- Secured **Top 10** in TechAccelerate Hackathon (BITS Hyderabad)
- Launched **Prmpt**, a prompt optimization SaaS product.
- Solved **200+ DSA problems** on LeetCode